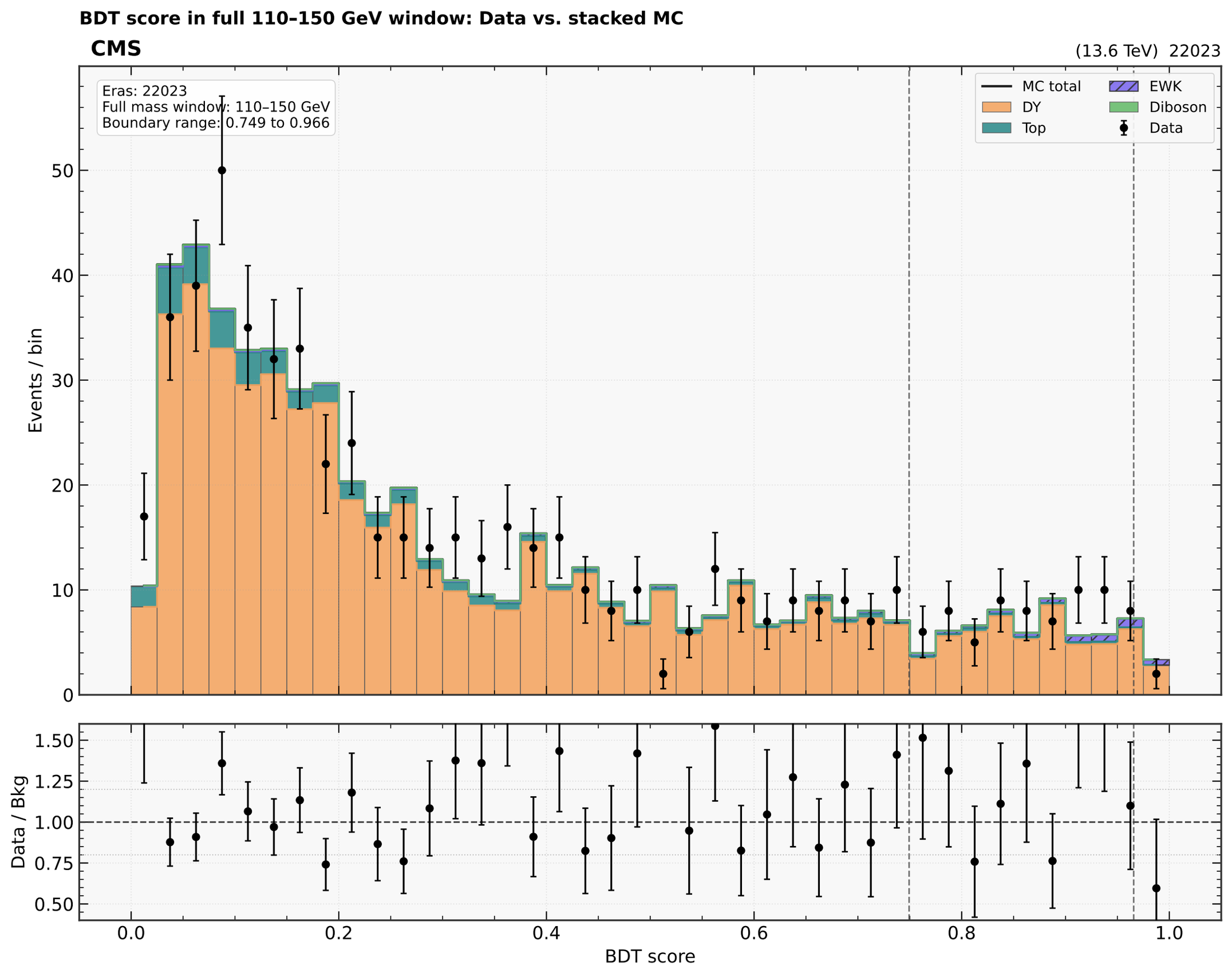
Run-3 H → μμ VBF BDT Category Optimisation

Eras : 22023
Category edges : 0.0000, 0.7493, 0.9356, 0.9656, 1.0000
MC scale factor : 1.0000 (Data/MC sideband normalisation)
Blinding window : 115–135 GeV
Sidebands : (110,115) ∪ (135,150) GeV
Outputs dir : /eos/home-m/memmad/HiggstoDimuon_VBF/src/drivers/9_TMVA/Outputs
Extra selection : –

CONTENTS:

p.2 BDT score distribution: Data vs. MC
p.3 BDT bin yields
p.4 Significance vs. number of categories
p.5 Boundary scan step-by-step (AN Fig. 32)
p.6 Atanh BDT + S/√B per category
p.7 BDT output vs. mass hypothesis (AN Fig. 26R)
p.8 Signal FWHM vs. BDT score
p.9 ROC curve with AUC
p.10 Blinded dimuon mass spectrum
p.11 Mass sculpting check (AN Fig. 25)
p.12 Per-category mass distributions and background fits
p.13 Background model comparison (χ²/ndf)
p.14 VBF signal DCB fits per category
p.15 DCB parameter table
p.16 σ_CB vs. category
p.17 Signal acceptance × efficiency
p.18 Per-category yield and sensitivity summary
p.19 Signal mass shapes mH=120/125/130 (AN Fig. 40)
p.20 Category summary table
p.21 Category process composition

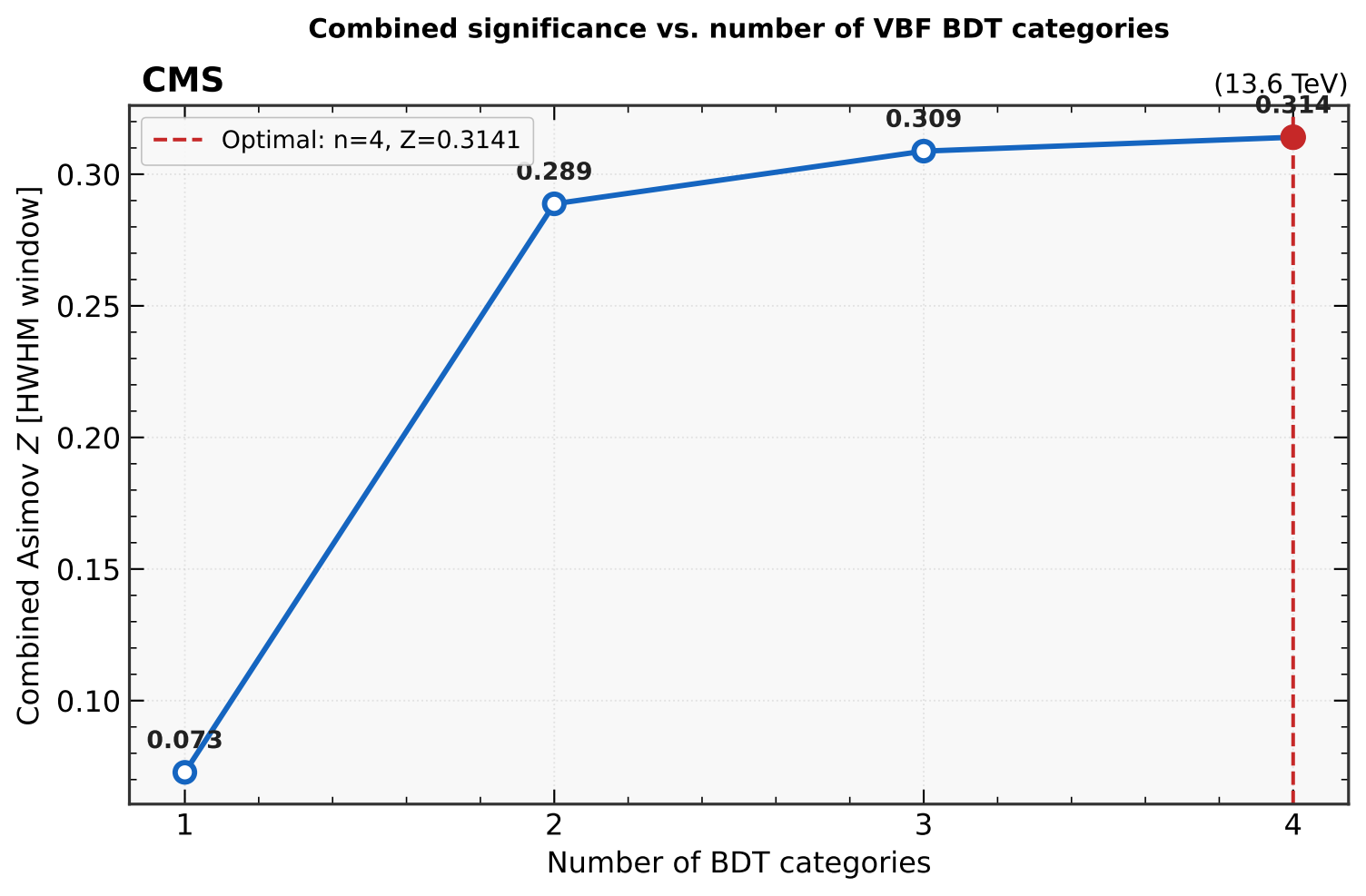

BDT score in full 110–150 GeV window: Data vs. stacked MC
CMS
(13.6 TeV) 22023
Eras: 22023
Full mass window: 110–150 GeV
Boundary range: 0.749 to 0.966
MC total
DY
Top
EWK
Diboson
Data
Events / bin
Data / Bkg
BDT score

BDT bin summary: signal, background, and sensitivity (page 1 / 2)

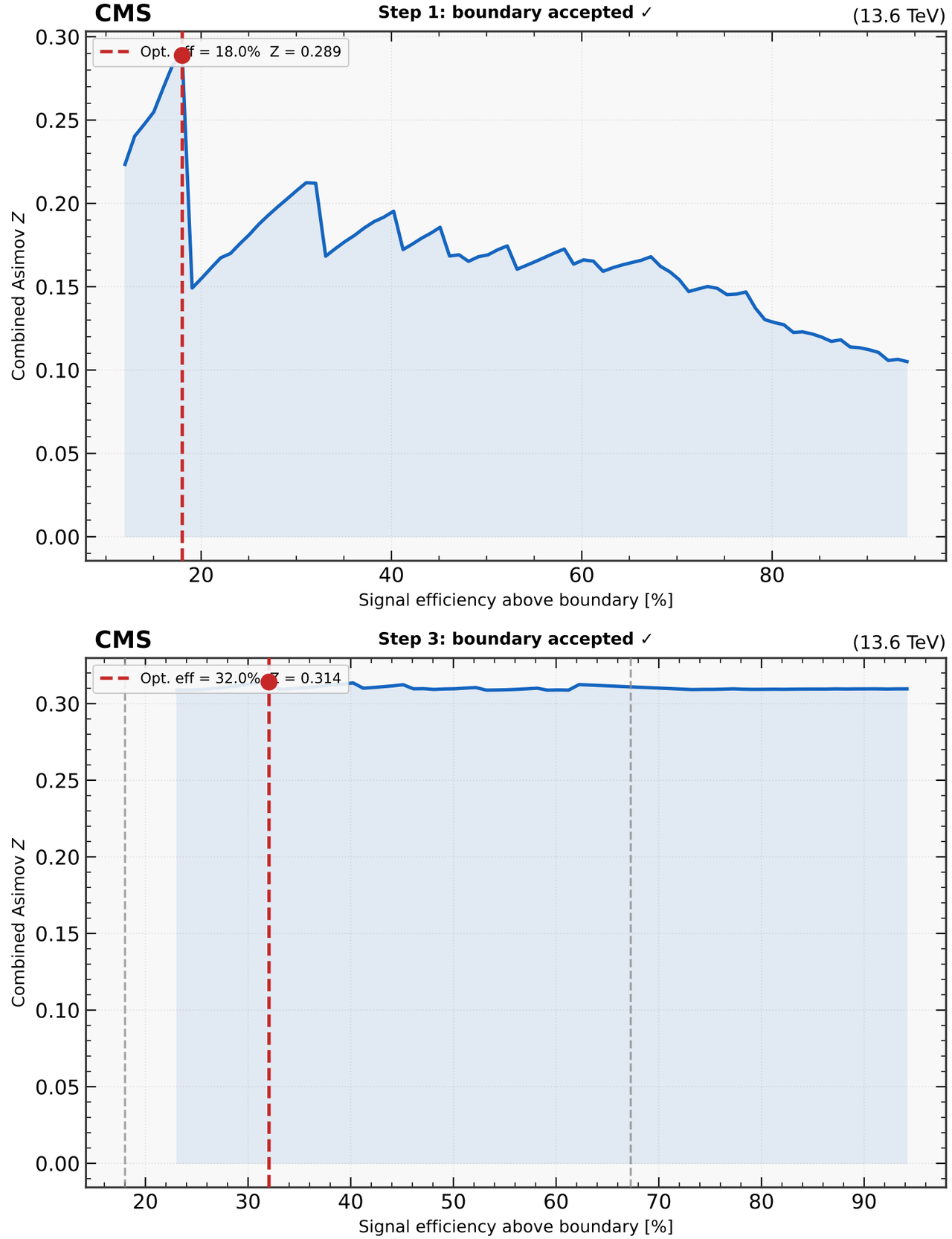
BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.000, 0.025)	0.001	10.39 ± 1.80	17	0.00
[0.025, 0.050)	0.002	41.03 ± 3.80	36	0.01
[0.050, 0.075)	0.005	42.90 ± 4.05	39	0.01
[0.075, 0.100)	0.005	36.79 ± 3.81	50	0.01
[0.100, 0.125)	0.006	32.85 ± 3.62	35	0.02
[0.125, 0.150)	0.007	32.99 ± 3.64	32	0.02
[0.150, 0.175)	0.008	29.10 ± 3.45	33	0.03
[0.175, 0.200)	0.008	29.69 ± 3.48	22	0.03
[0.200, 0.225)	0.009	20.34 ± 2.80	24	0.05
[0.225, 0.250)	0.009	17.33 ± 2.54	15	0.05
[0.250, 0.275)	0.009	19.73 ± 2.70	15	0.05
[0.275, 0.300)	0.011	12.92 ± 2.16	14	0.08
[0.300, 0.325)	0.011	10.90 ± 2.16	15	0.10
[0.325, 0.350)	0.013	9.56 ± 1.87	13	0.13
[0.350, 0.375)	0.012	8.93 ± 1.81	16	0.14
[0.375, 0.400)	0.012	15.38 ± 2.57	14	0.08
[0.400, 0.425)	0.012	10.46 ± 2.09	15	0.12
[0.425, 0.450)	0.013	12.13 ± 2.30	10	0.11
[0.450, 0.475)	0.015	8.86 ± 1.85	8	0.17
[0.475, 0.500)	0.015	7.05 ± 1.68	10	0.21

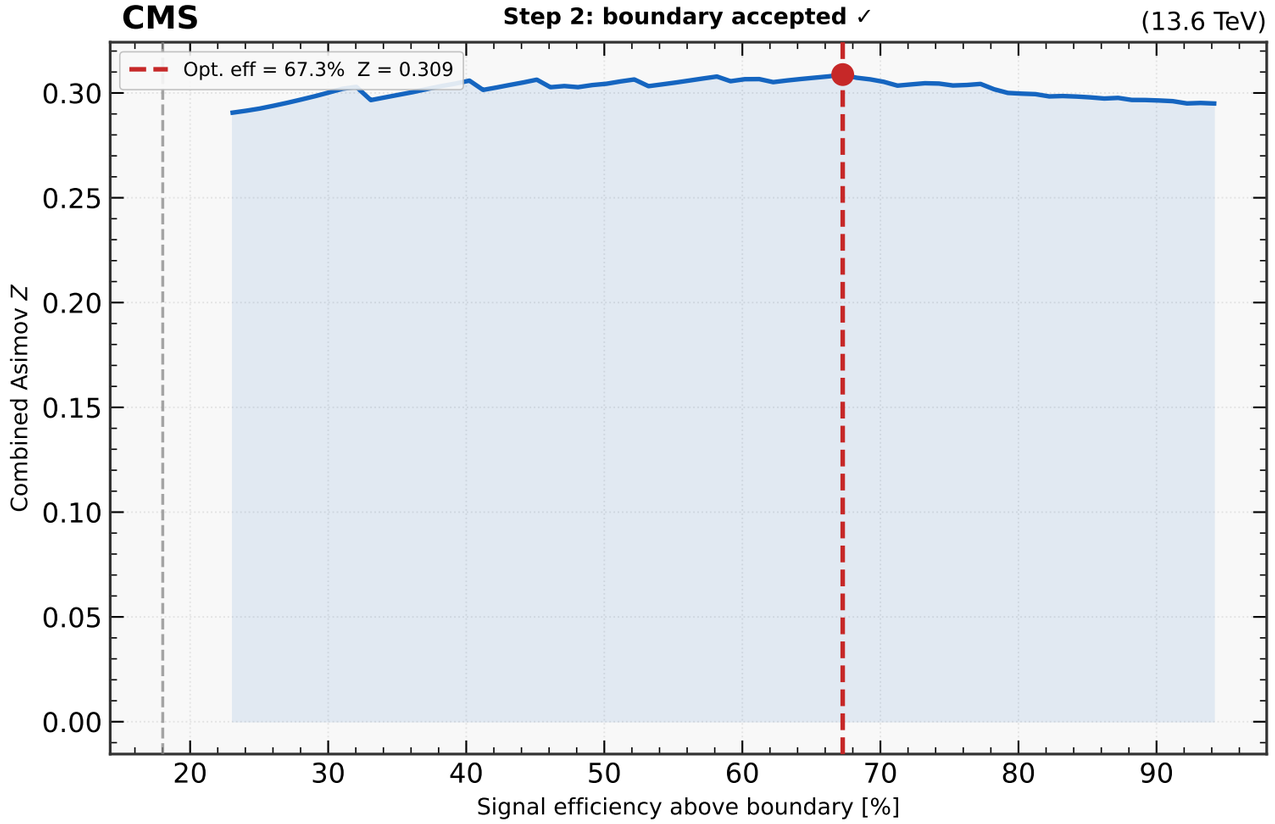
BDT bin summary: signal, background, and sensitivity (page 2 / 2)

BDT bin	Total signal	Bkg. ± ΔB	Data	S/(S+B) (%)
[0.500, 0.525)	0.015	10.44 ± 2.06	2	0.15
[0.525, 0.550)	0.017	6.33 ± 1.47	6	0.26
[0.550, 0.575)	0.017	7.56 ± 1.66	12	0.23
[0.575, 0.600)	0.019	10.90 ± 2.06	9	0.18
[0.600, 0.625)	0.020	6.69 ± 1.57	7	0.30
[0.625, 0.650)	0.021	7.06 ± 1.83	9	0.30
[0.650, 0.675)	0.023	9.48 ± 2.10	8	0.24
[0.675, 0.700)	0.025	7.32 ± 1.67	9	0.34
[0.700, 0.725)	0.027	8.00 ± 1.89	7	0.33
[0.725, 0.750)	0.028	7.09 ± 1.61	10	0.40
[0.750, 0.775)	0.034	3.96 ± 1.19	6	0.85
[0.775, 0.800)	0.037	6.09 ± 1.57	8	0.61
[0.800, 0.825)	0.043	6.60 ± 1.70	5	0.64
[0.825, 0.850)	0.048	8.10 ± 1.81	9	0.59
[0.850, 0.875)	0.057	5.89 ± 1.53	8	0.96
[0.875, 0.900)	0.070	9.18 ± 1.90	7	0.76
[0.900, 0.925)	0.091	5.65 ± 1.42	10	1.59
[0.925, 0.950)	0.119	5.76 ± 1.42	10	2.03
[0.950, 0.975)	0.169	7.27 ± 1.62	8	2.27
[0.975, 1.000)	0.148	3.36 ± 1.12	2	4.23

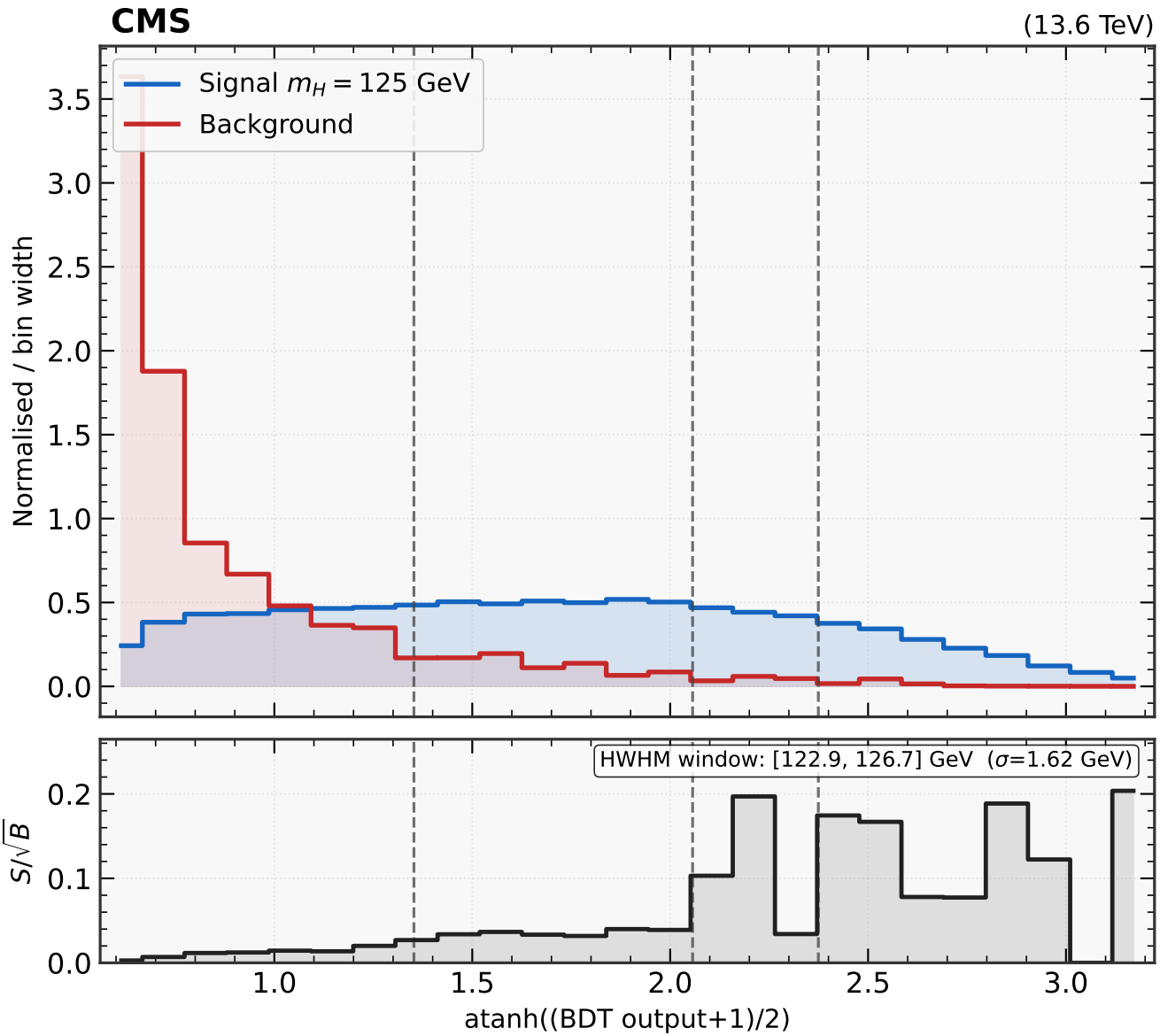

Combined significance vs. number of VBF BDT categories
CMS
(13.6 TeV)
Optimal: n=4, Z=0.3141
0.073
0.289
0.309
0.314
Combined Asimov Z [HWHM window]
0.10
0.15
0.20
0.25
0.30
1
2
3
4
Number of BDT categories

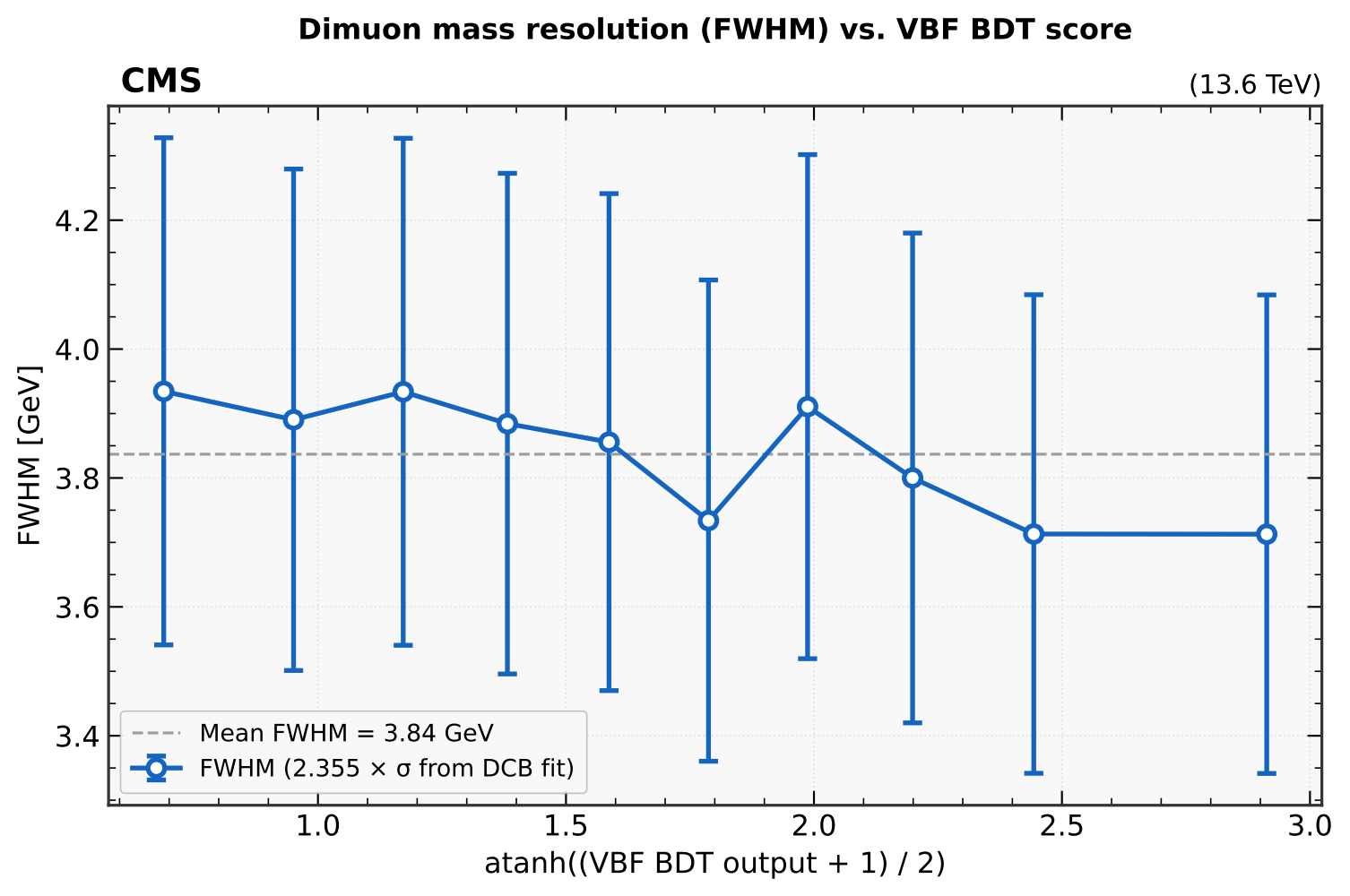
Iterative category boundary optimisation (AN Fig. 32 analog)

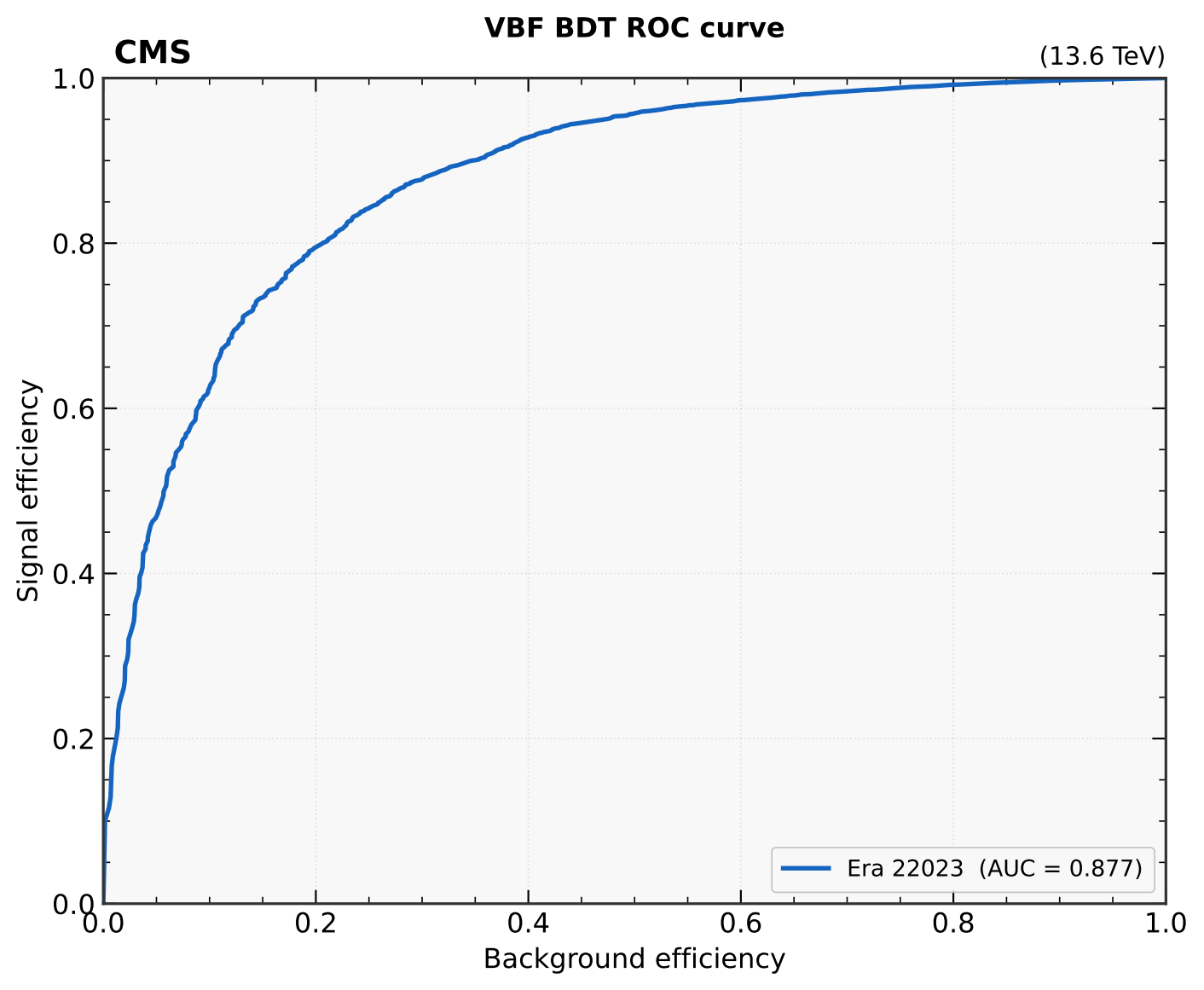


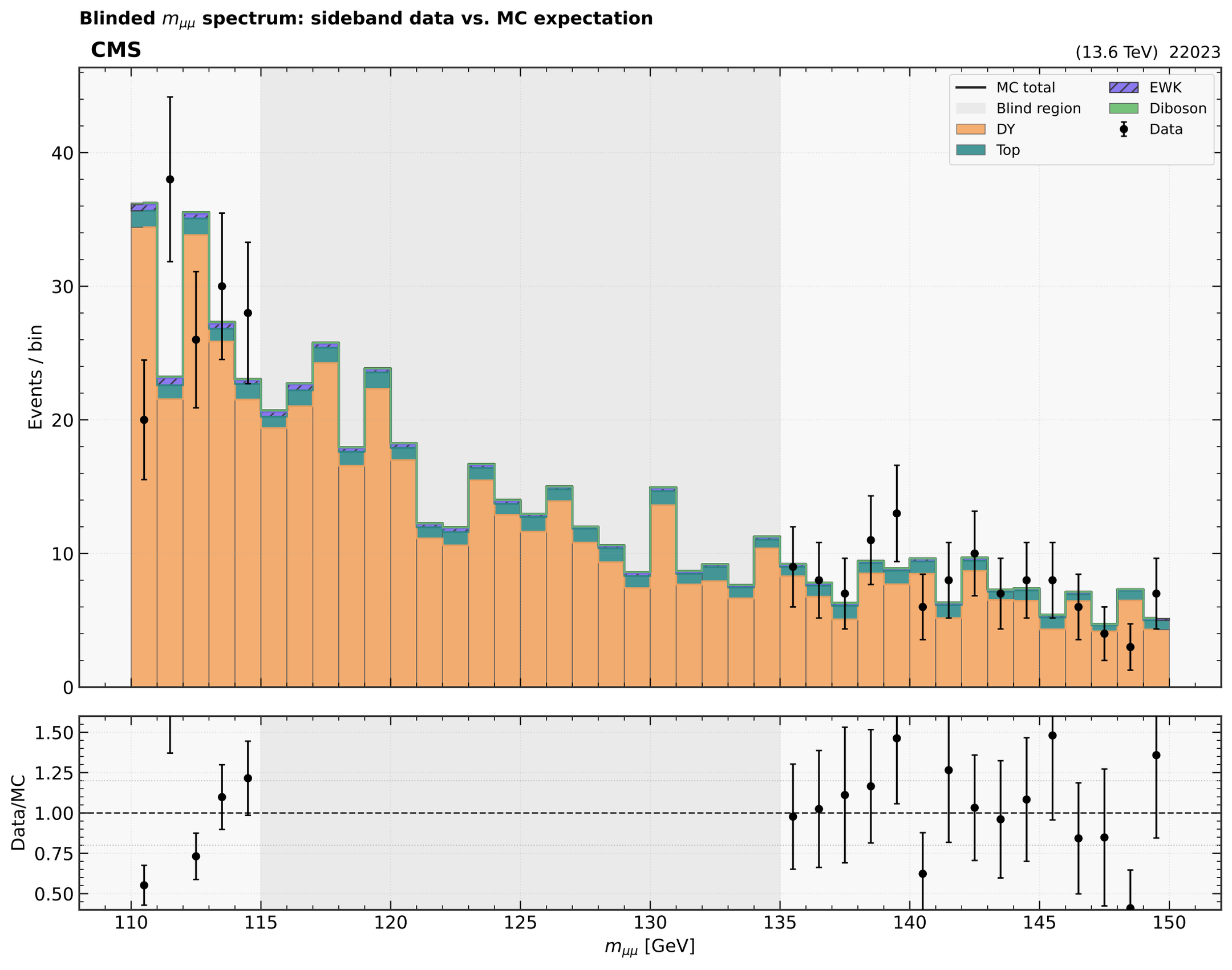


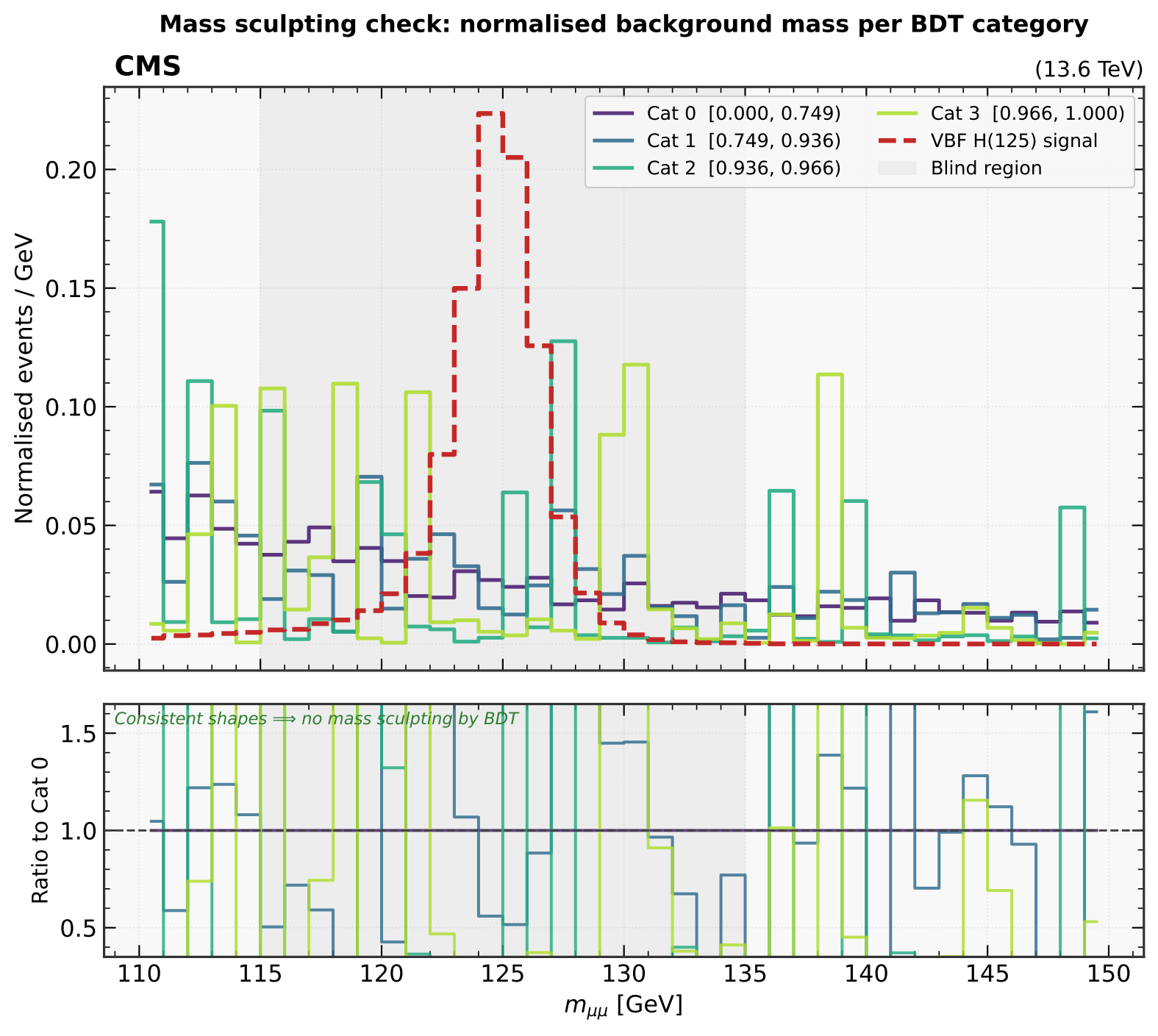
VBF BDT score and per-bin $S/\sqrt{B}$ in signal HWHM window



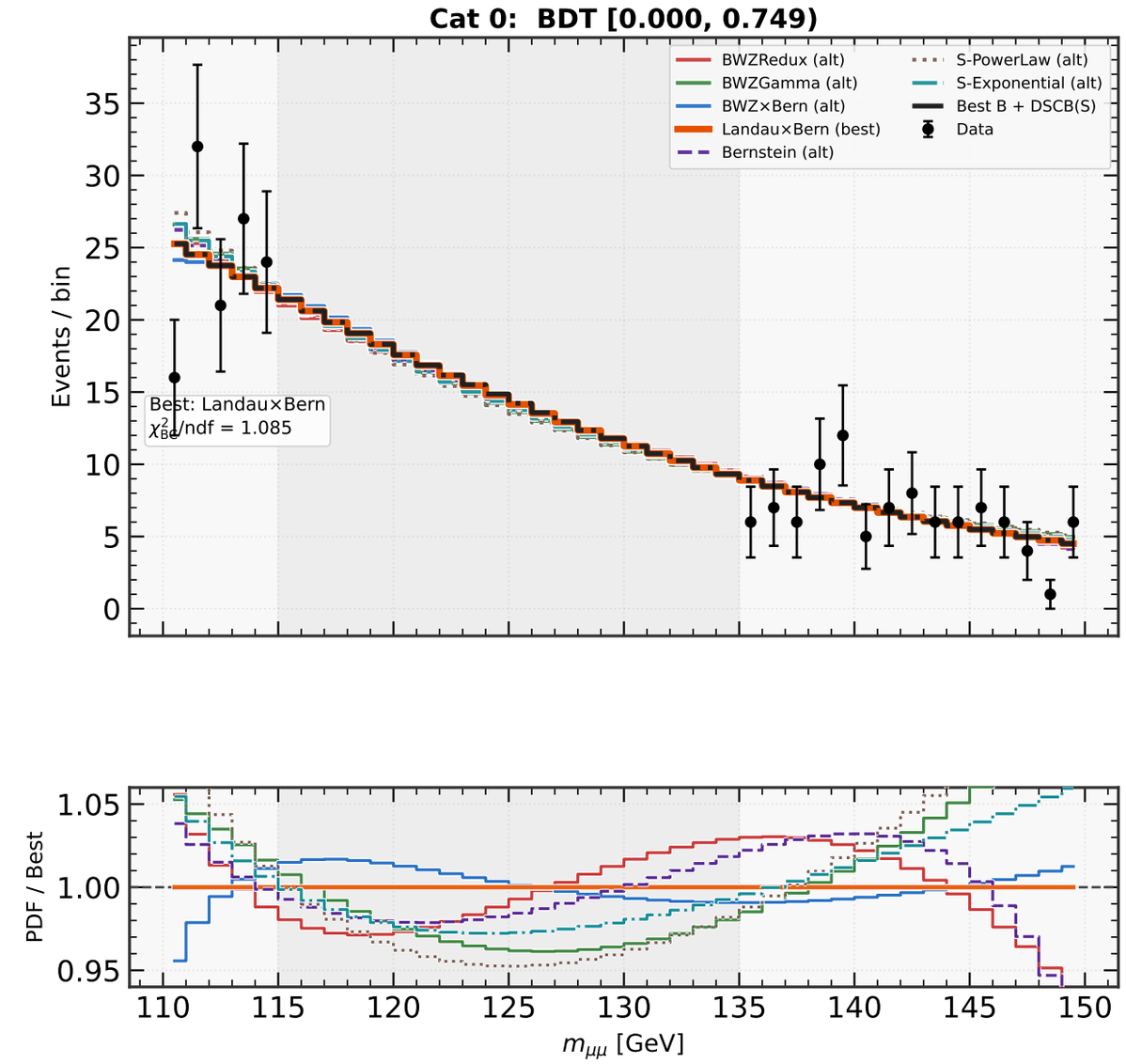

Dimuon mass resolution (FWHM) vs. VBF BDT score
CMS
(13.6 TeV)
FWHM [GeV]
4.2
4.0
3.8
3.6
3.4
1.0
1.5
2.0
2.5
3.0
atanh((VBF BDT output + 1) / 2)
Mean FWHM = 3.84 GeV
FWHM (2.355 × σ from DCB fit)

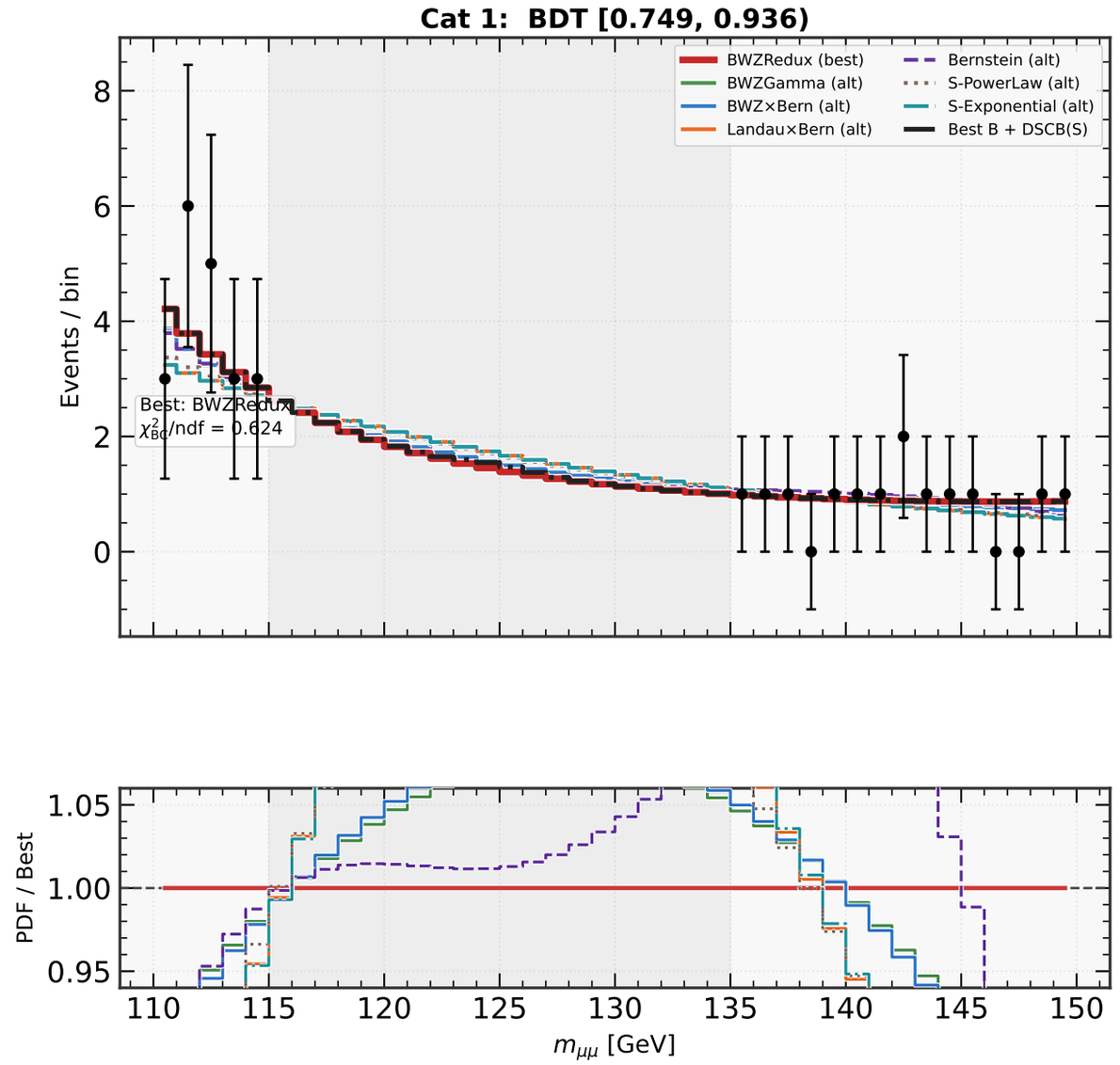

VBF BDT ROC curve
CMS
(13.6 TeV)
Signal efficiency
Background efficiency
0.0
0.2
0.4
0.6
0.8
1.0
Era 22023 (AUC = 0.877)

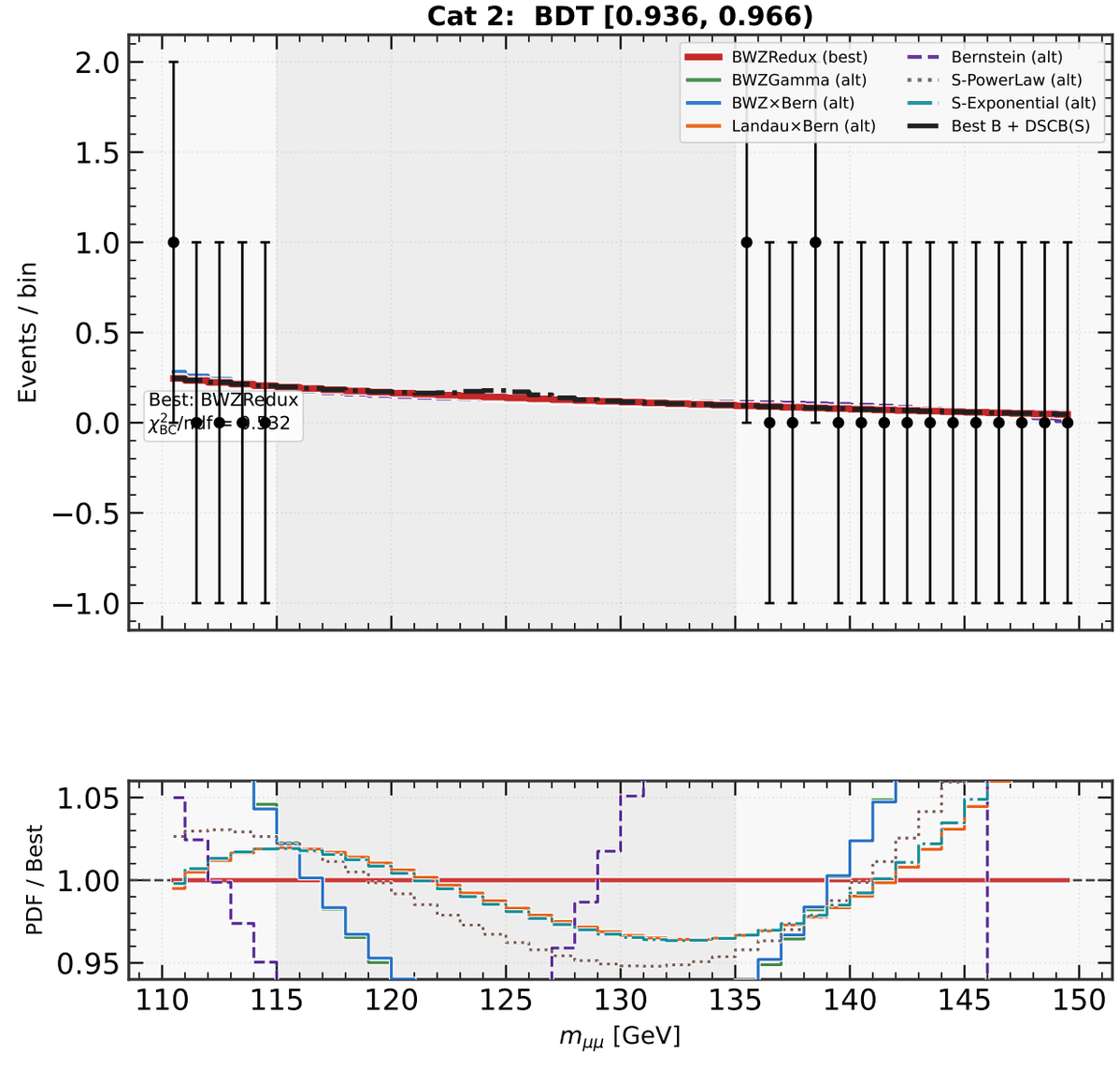

Blinded $m_{\mu\mu}$ spectrum: sideband data vs. MC expectation
CMS
(13.6 TeV) 22023
MC total
Blind region
DY
Top
EWK
Diboson
Data
Events / bin
Data/MC
$m_{\mu\mu}$ [GeV]

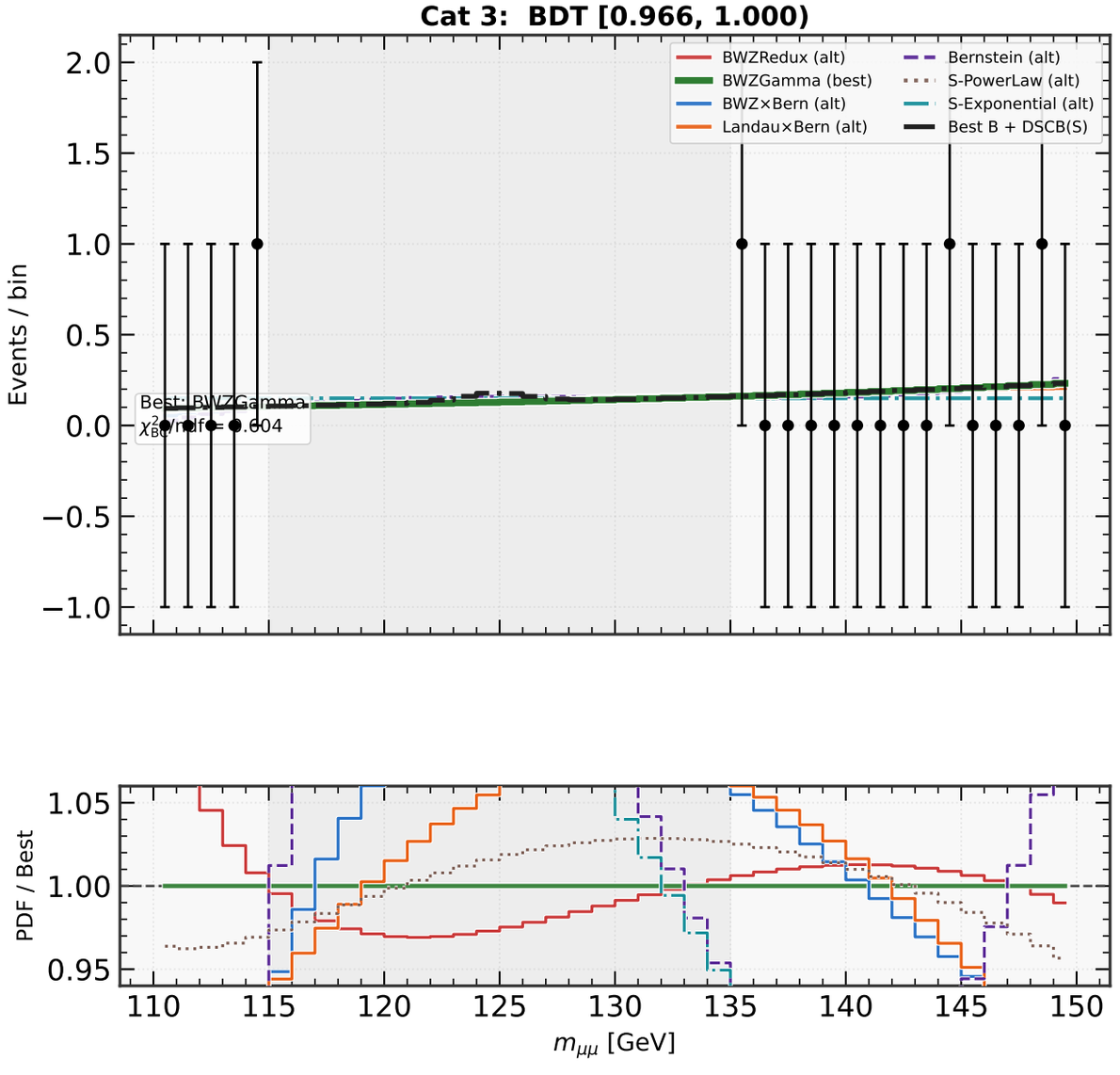

Mass sculpting check: normalised background mass per BDT category
CMS
(13.6 TeV)
Cat 0 [0.000, 0.749)
Cat 1 [0.749, 0.936)
Cat 2 [0.936, 0.966)
Cat 3 [0.966, 1.000)
VBF H(125) signal
Blind region
Normalised events / GeV
0.00
0.05
0.10
0.15
0.20
Consistent shapes ⟹ no mass sculpting by BDT
Ratio to Cat 0
0.5
1.0
1.5
110
115
120
125
130
135
140
145
150
$m_{\mu\mu}$ [GeV]

Per-category dimuon mass: background model fits





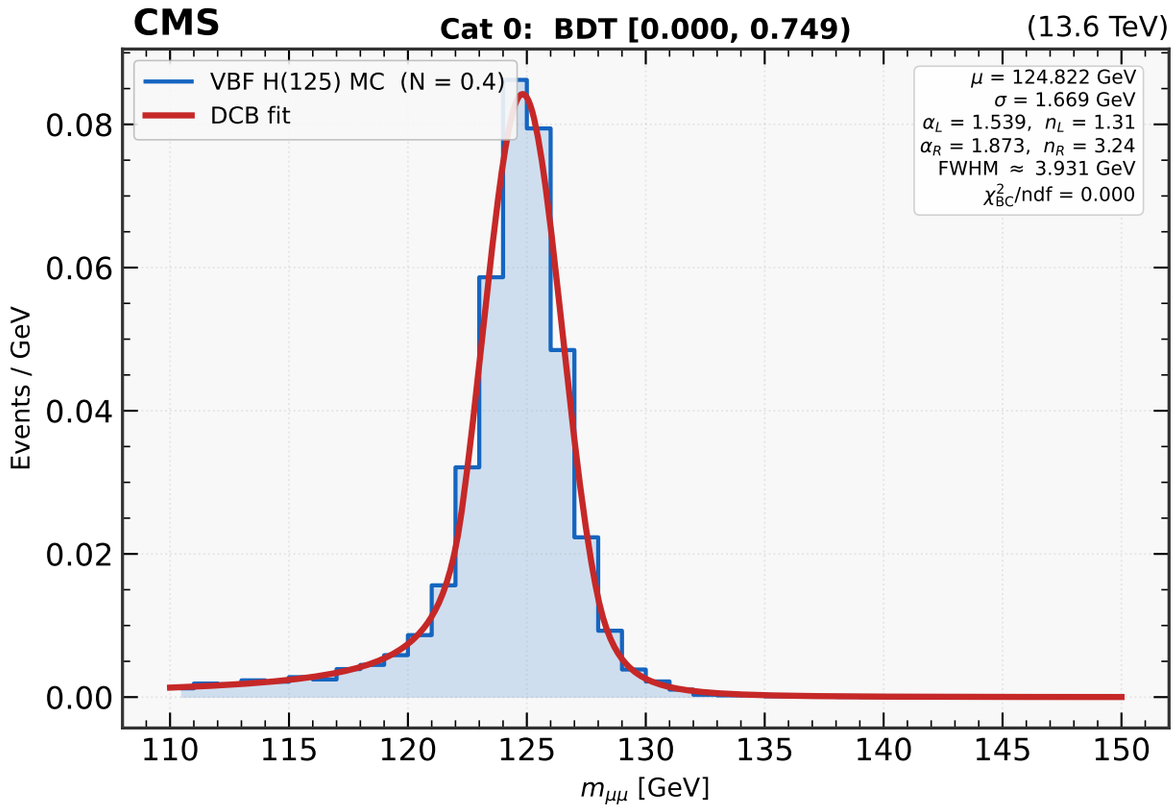


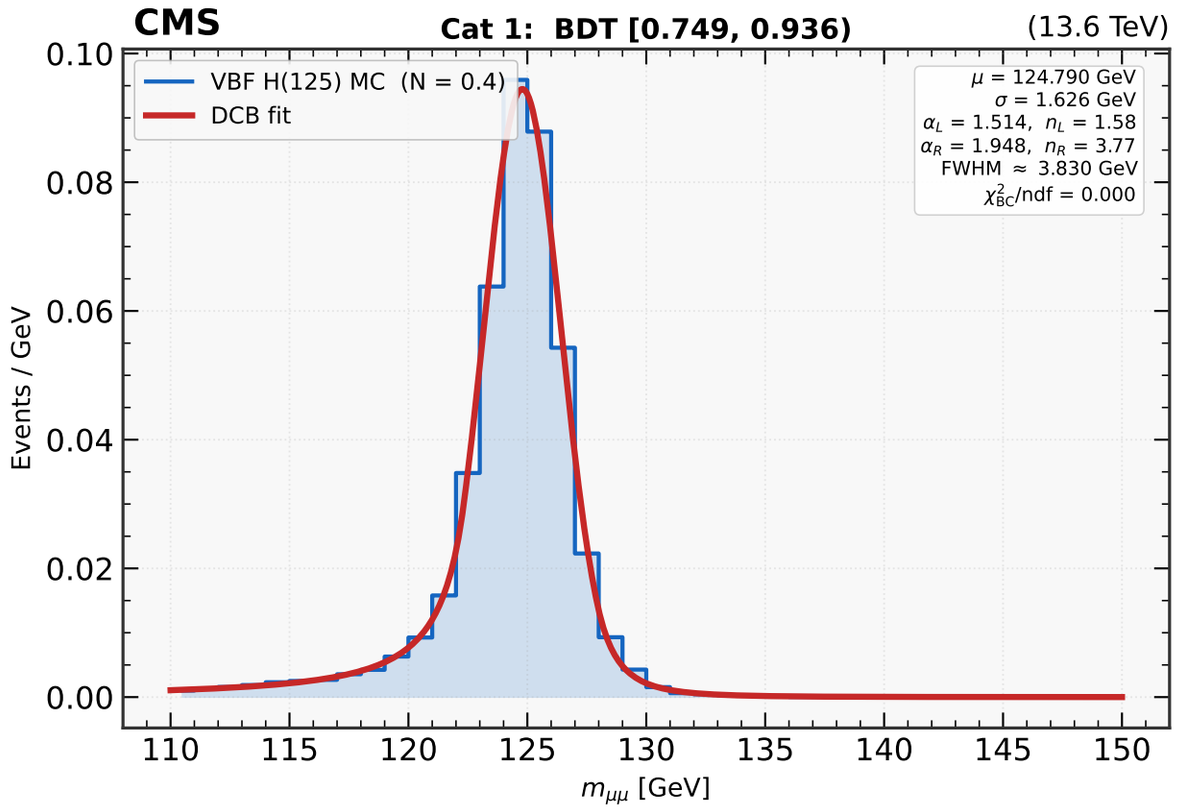


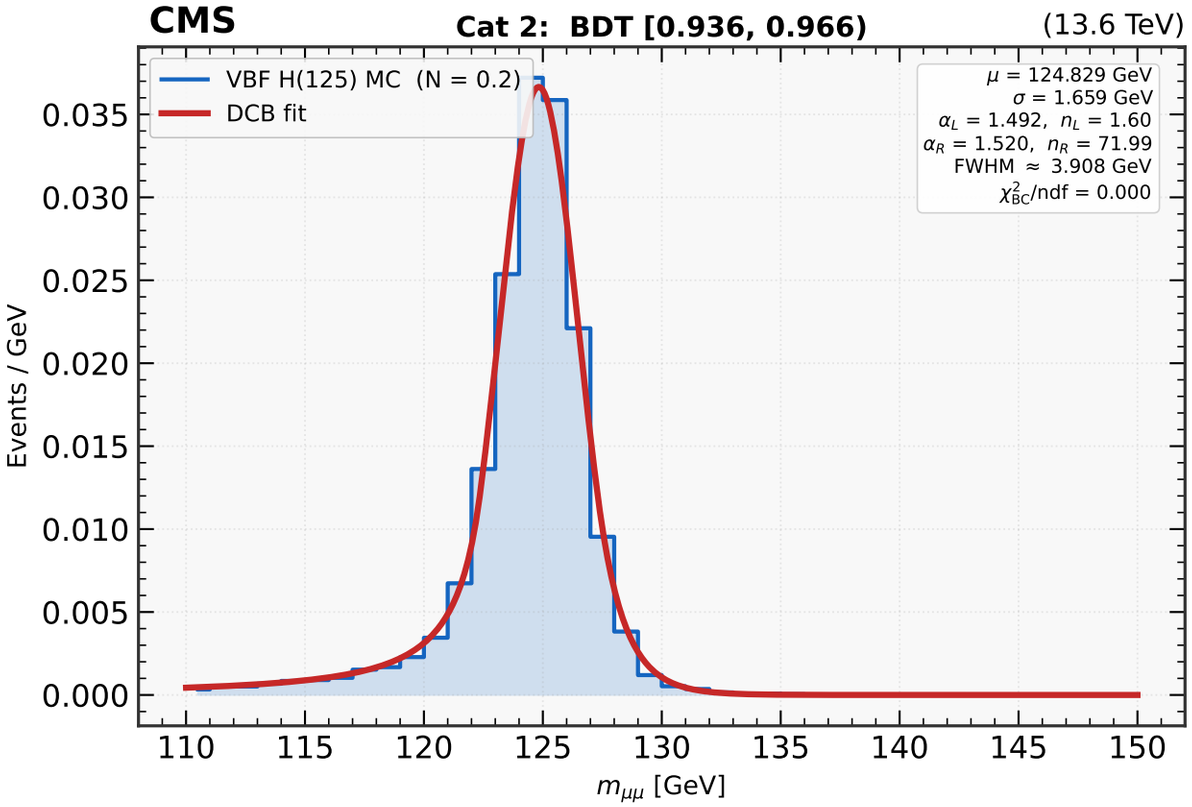
Per-category background model fit comparison (χ^2_BC/ndf, ndf)

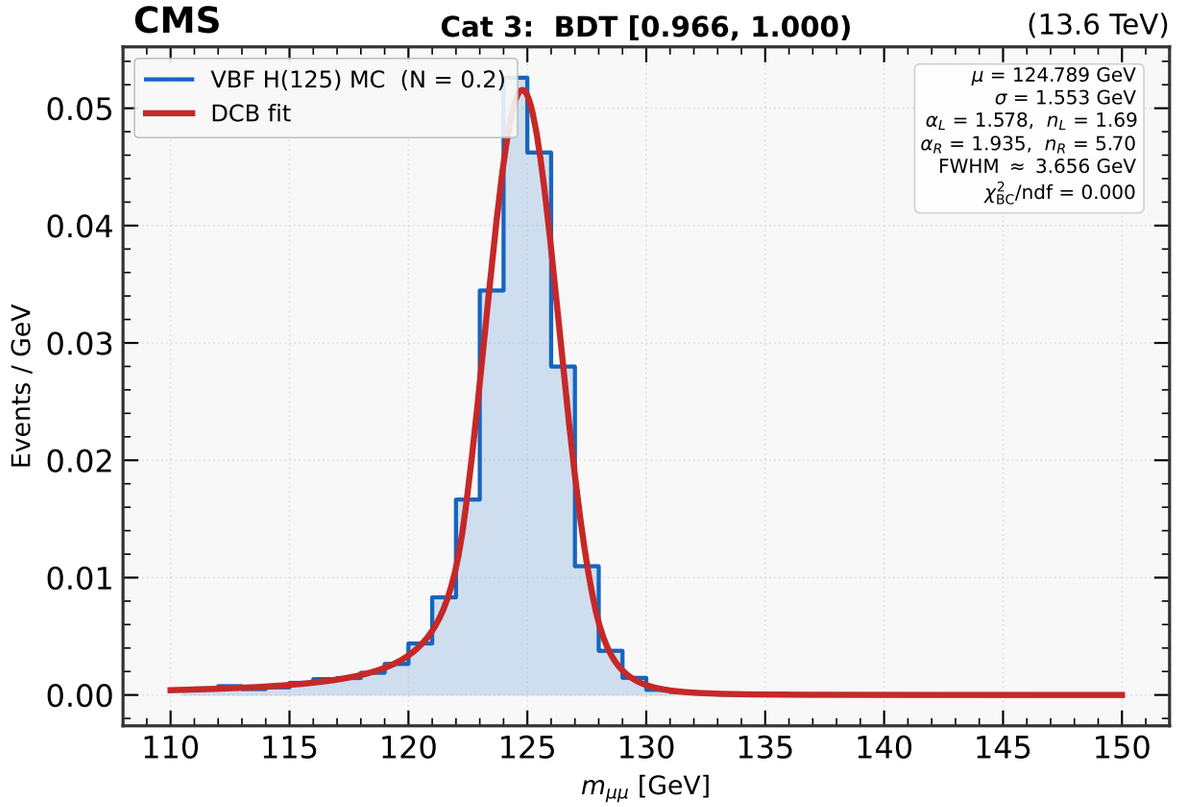
Category	BDT range	Best model	Best χ^2_BC/ndf	BWZRedux	BWZGamma	BWZ×Bern	Landau×Bern	Bernstein	S-PowerLaw	S-Exponential
0	[0.000, 0.749)	Landau×Bern	1.085	1.098 (37)	1.095 (37)	1.105 (36)	1.085 (37)	1.147 (35)	1.111 (37)	1.094 (37)
1	[0.749, 0.936)	BWZRedux	0.624	0.624 (37)	0.630 (37)	0.649 (36)	0.677 (37)	0.663 (35)	0.665 (37)	0.676 (37)
2	[0.936, 0.966)	BWZRedux	0.532	0.532 (37)	0.534 (37)	0.549 (36)	0.535 (37)	0.545 (35)	0.535 (37)	0.535 (37)
3	[0.966, 1.000)	BWZGamma	0.604	0.605 (37)	0.604 (37)	0.620 (36)	0.605 (37)	0.632 (35)	0.604 (37)	0.615 (37)

VBF H(125) signal DCB extraction per BDT category



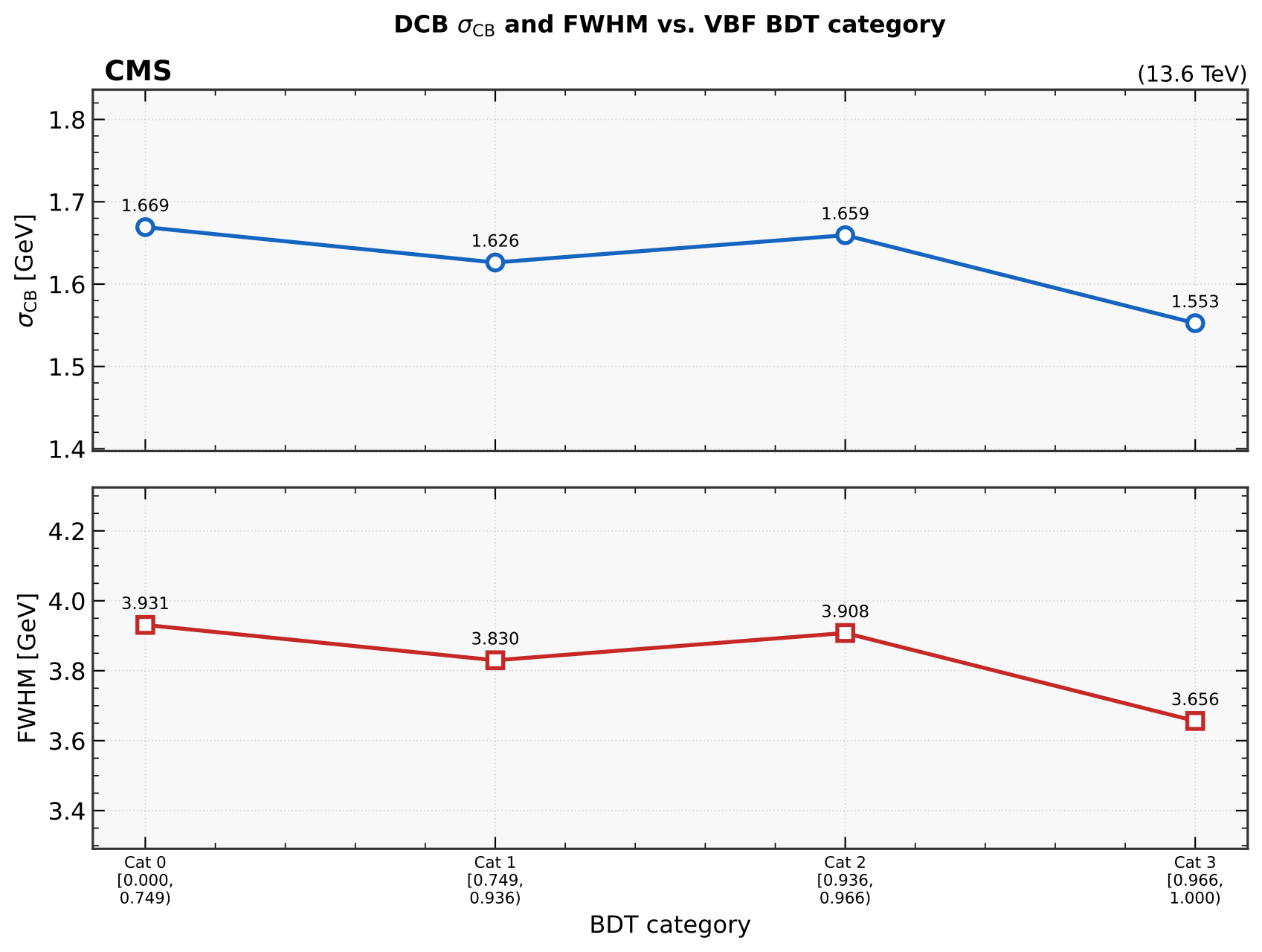


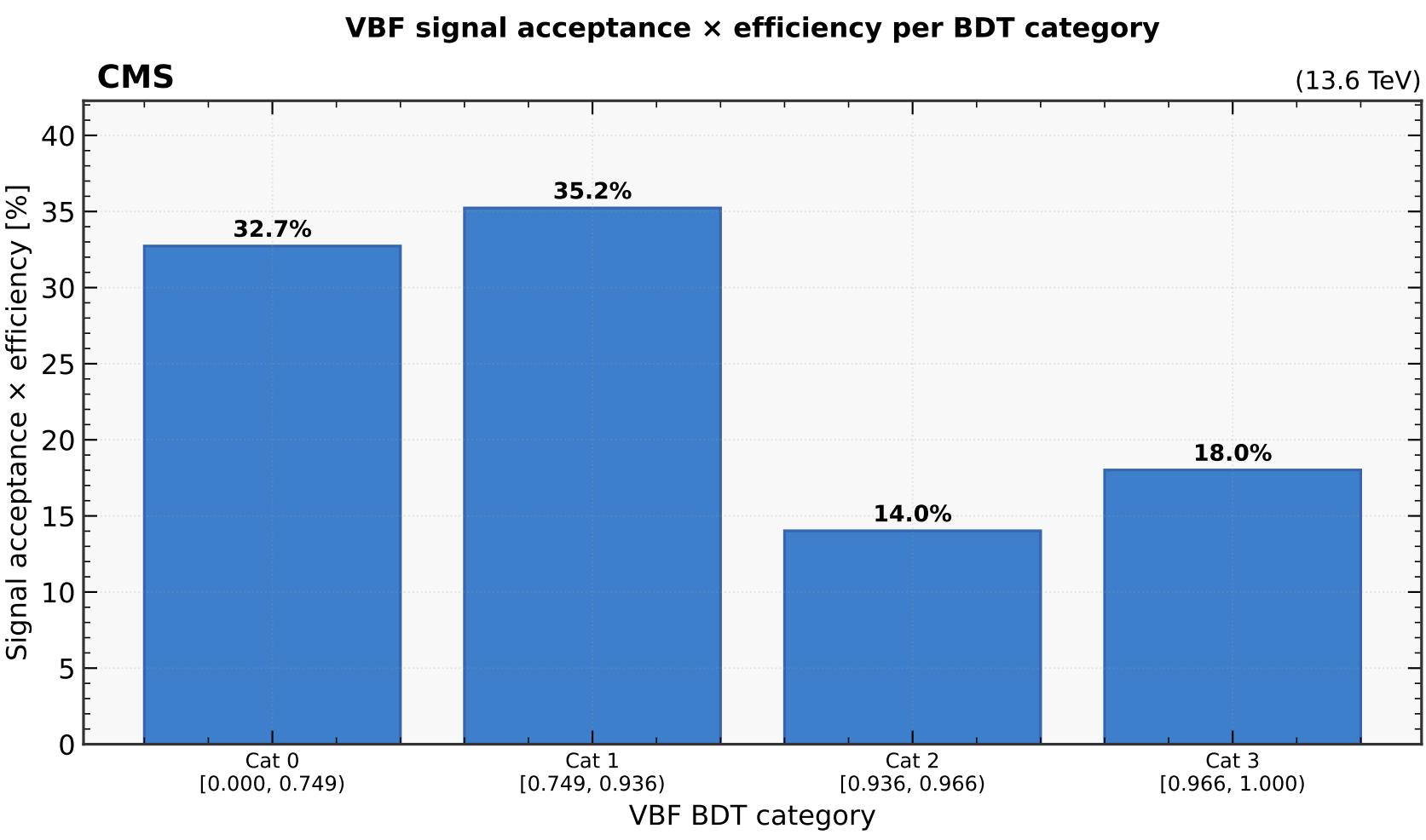




VBF signal DCB fit parameters per category

Category	BDT range	N_sig	Mean [GeV]	Sigma [GeV]	alpha_L	n_L	alpha_R	n_R	FWHM [GeV]	Asimov Z	χ^2_BC/ndf
0	[0.000, 0.749)	0.40	124.8224	1.6692	1.5392	1.312	1.8733	3.240	3.931	0.024	0.000
1	[0.749, 0.936)	0.43	124.7904	1.6263	1.5138	1.577	1.9485	3.773	3.830	0.093	0.000
2	[0.936, 0.966)	0.17	124.8294	1.6594	1.4924	1.599	1.5201	71.987	3.908	0.098	0.000
3	[0.966, 1.000)	0.22	124.7887	1.5526	1.5783	1.690	1.9353	5.705	3.656	0.285	0.000


DCB σ_{CB} and FWHM vs. VBF BDT category
CMS
(13.6 TeV)
σ_{CB} [GeV]
1.8
1.7
1.6
1.5
1.4
1.669
1.626
1.659
1.553
FWHM [GeV]
4.2
4.0
3.8
3.6
3.4
3.931
3.830
3.908
3.656
Cat 0 [0.000, 0.749)
Cat 1 [0.749, 0.936)
Cat 2 [0.936, 0.966)
Cat 3 [0.966, 1.000)
BDT category


VBF signal acceptance × efficiency per BDT category
CMS
(13.6 TeV)
Signal acceptance × efficiency [%]
0
5
10
15
20
25
30
35
40
32.7%
35.2%
14.0%
18.0%
Cat 0
[0.000, 0.749)
Cat 1
[0.749, 0.936)
Cat 2
[0.936, 0.966)
Cat 3
[0.966, 1.000)
VBF BDT category

Per-category yield and sensitivity summary

Category	BDT range	S_full	B_full	S_HWHM	B_HWHM	HWHM window [GeV]	S/√B	Asimov Z
0	[0.000, 0.749)	0.40	1173.46	0.272	126.274	[122.9, 126.8]	0.024	0.024
1	[0.749, 0.936)	0.43	116.76	0.295	9.880	[122.9, 126.7]	0.094	0.093
2	[0.936, 0.966)	0.17	19.44	0.119	1.429	[122.9, 126.8]	0.100	0.098
3 ★	[0.966, 1.000)	0.22	11.95	0.154	0.243	[123.0, 126.6]	0.311	0.285

Category summary: HWHM-windowed signal and background yields

Category	BDT range	Data sidebands	MC sidebands ± stat	Data/MC (sidebands)	MC full (110-150) ± stat	S/√B [HWHM]	Asimov Z (cat) [HWHM]	B_HWHM
0	[0.000, 0.749)	217	545.70 ± 22.68	0.398	1173.46 ± 33.17	0.0242	0.0242	126.274
1	[0.749, 0.936)	33	55.15 ± 7.16	0.598	116.76 ± 10.51	0.0938	0.0934	9.880
2	[0.936, 0.966)	3	10.35 ± 3.15	0.290	19.44 ± 4.14	0.0998	0.0985	1.429
3 ★	[0.966, 1.000)	4	4.04 ± 1.73	0.989	11.95 ± 3.09	0.3114	0.2851	0.243
Combined	all						0.3167	

Category composition by process (full mass window)

Category	BDT range	DY	EWK	Top	Diboson	total	total_err2
0	[0.000, 0.749)	1075.56	10.76	82.88	4.26	1173.46	1099.94
1	[0.749, 0.936)	105.44	6.01	4.63	0.68	116.76	110.39
2	[0.936, 0.966)	16.52	2.27	0.58	0.07	19.44	17.17
3	[0.966, 1.000)	9.78	1.86	0.30	0.01	11.95	9.54